PROCESSING OF THE DENTURE

(FLASKING)

Flasking of the denture: is the process of investing the cast with the waxed denture in a flask to make a two sectional mold used to form the acrylic resin denture base.

The flask: is a metal case or tube used in investing procedures or can defined as a sectional metal case in which a sectional mold is made of artificial stone or plaster of Paris for the purpose of compressing and processing dentures or other resinous restorations.

The flask is consists of:

1. Lower half that contains the cast.
2. Upper half.
3. The cover or lid.

Flasking techniques:

1- Compression technique (open- pack method).

The master cast with the waxed-trial denture invested in the lower half of the flask with gypsum investment material. Then the upper half of the flask put in place and gypsum poured to the occlusal surfaces of the teeth. The top portion of the flask poured with another layer of plaster or stone and the cover placed on the flask.

2- Injection molding technique:

It is a complicated procedure required special flask and equipment .In this technique the wax pattern is sprued and the material injected into the mold. This process allows injection of further material during polymerization to compensate for the polymerization shrinkage.

The procedure of processing the denture in compression technique:

1. Sealing of the upper and lower trial dentures all over the border until the margin of the cast while the casts on the articulator.
2. Soak the cast and the mounting plaster in water a few minute to separate the cast from the mounting plaster, save the plaster mounting, as it will be used to reposition the cast on the articulator after the dentures are processed.
3. Make sure that there is enough space between the incisal and occlusal surface of the teeth and the top of the upper ring about 3-6 mm., if there is no space then the cast must be reduced in thickness
4. Adapt a layer of tinfoil to the base of the casts, slightly overlapping the edges to insure clean removal from the investment, apply separating medium to the cast and flask.

5. Invest the lower half of the flask first:

a. Use a mixture of plaster or stone and placed in the base of the flask.

b. Center the cast in the lower half of the flask and pushed to place until the bottom of the cast touches the base of the flask. Note that the posterior portion of the cast is level with the edge of the flask.

c. Remove any undercuts in the plaster. Undercuts will prevent the separation of the upper ring from the lower portions of the flask.

d. The plaster smoothed even with the base of the cast and allows the plaster to set.

e. After the final setting of plaster has occurred, it will be coated with separating medium.

6. Investing the upper half of the flask:

a. Reposition the ring (upper portion) of the flask.

b. Mixing of stone or plaster done, pouring of the plaster to flow and reach all surfaces of teeth without any air babbles by putting the flask on the vibrator, the plaster or stone must reach the incisal edge and occlusal surface of teeth.

c. Separating medium painted on the second layer of plaster after setting and smoothing of plaster.

d. A third mixing of stone done and pouring it until the flask is filled with stone and the flask is covered and some stone enter through the holes in the cover and around the edges, it is essential to have metal-to-metal contact.

e. The stone left to set completely for about 45 minutes.

7. Wax elimination

After the complete set of the gypsum, the flask is ready for the next step which is "Wax elimination":

a. Place the flask on a ladle and lower it into boiling water for 5 minutes. This will soften the waxed denture base, which can easily remove from the mold when the flask opened.

b. After five min., remove the flask from the boiling water and gently open it, insert a wax knife between the lower and upper half and gently separate them.

c. Remove the semisolid pieces of the waxed denture base. All the teeth should remain in the top half of the flask. Using more hot (boiling) water to flush out all the remaining wax.

d. Wax solvent can be used with stiff brush to remove any remaining wax on teeth.

e. As soon as possible, flush the mold with clean hot water to which a detergent has been added. The detergent will be flush out the wax residue from area that cannot reached with the wax solvent. Immediately flush the mold with hot water to remove all traces of the detergent solution.

f. It is essential to remove all wax residue. Acrylic resin will not adhere to a surface coated with wax.

g. Stand the flask on its side and allow it to drain, dried and cooled.

8. Packing and Curing:

a. Separating medium used on plaster or stone, care should be taken not to painting the teeth with separating medium.

b. The flask is left to dry and another coat is painted on the flask and left to dry.

c. Hot or heat cure acrylic is used polymer/ monomer mixed according to manufacture instruction. Usually 10 CC of monomer and 30 CC of polymer will be enough to pack an average-sized denture, after mixing of the material on clean jar and reach dough stage, it's ready for packing.

d. Pack the material in the upper half of the flask, being sure to press it well into the area around the teeth. Use enough material to insure over-packing on the first closure using nylon sheet. At least two trial closures done and before the final closure, a thin layer-separating medium applied on the cast and the nylon pressure by bench hydraulic press of about 100 kg/cm2. Then the flask is put in spring clamp sheet is removed and then the two halves of the flask are closed and the clamp is closed tightly, it is essential to have metal-to-metal contact.

9. Curing :

It is the process of polymerization of acrylic resin by heat, the amount of heat must controlled while processing acrylic resin**.**

Types of curing cycle:

A. ***Slow curing cycle:*** The flask is heat to 70ºC for 7 hours then 100 ºC for 3 hours (the total 10 hours), or heated to 75 ºC for 6 hours then 100 ºC for 1 hour or 74 ºC for 8 hours.

B. ***Rapid curing cycle:*** The flask is heated to 140 ºC for10 minutes with 40 P.S.I. pressure.

The best curing cycle is the slow curing cycle because most of the conversion of monomer to polymer occurs during the period at 70 ºC and the rapid cycle method may induce greater dimensional changes in the dentures than slow – cure method.

10. Deflasking:

It is the removal of the mold from the flask and separate the mold from the denture and the cast. The flask removed from the mold using a flask ejector, which used to separate the flask from the mold after removing of the cover. By using a saw, longitudinal and horizontal cuts carefully made through the plaster or stone and the pieces gently removed. The cured dentures and their casts have been removed from the mold.

OCCLUSAL CORRECTION

Artificial teeth move about to a minor degree during waxing and processing of the trial denture to a resin one. This tooth movement is due primarily to dimensional changes in the wax denture base, the investing materials, and in the resin denture base during curing. Occlusal discrepancies caused by these dimensional changes ordinarily are removed before the dentures are polished.

Causes of errors in occlusions:

1. Inaccurate maxillo-mandibular relation record by the dentist.

2. Errors made in the transfer of maxillo-mandibular relation to the articulator.

3. Failure to use the face-bow and subsequently changing in the vertical relation on the articulator.

4. Failure to seat the occlusion rims correctly on the cast (ill-fitting record base).

5. Incorrect arrangement of the posterior teeth.

6. Failure to close the flask completely during processing.

7. Warpage of the dentures by overheating them in polishing stage.

8. Dimensional changes of the acrylic resin (denture base material).

The errors that are a result of processing changes can be eliminated before insertion of the dentures in the patient's mouth, correcting occlusal surface of the teeth by selective grinding.

Selective grinding:

Defined as "modification of the occlusal forms of the teeth by grinding according to a plan. The modification of the occlusal forms of teeth by grinding at selected places market by spots made by articulating paper, or marked by parts of the teeth cutting through a thin layer of wax placed over the teeth".

Correction of occlusal errors:

Two methods are generally used in correcting errors due to processing changes, In both, the occlusal surfaces of teeth are altered by selective grinding:

1. Intra-oral (inside the patient mouth).

2. Extra-oral (on the articulator in the laboratory).

Disadvantages of intra-oral method correction:

1. It is difficult to see the errors because the soft tissues will be distorted and obscure the errors.

2. The denture bases will be shift in relation to the underlying bone when there are errors in occlusion due to the resiliency of the soft tissues.

3. The articulating paper marks are likely to be incorrect due to the presence of the saliva.

4. The central of jaw position depends entirely on the ability of the patient to place and move jaw correctly.

Advantages of extra-oral (on the articulator) correction:

1. More visible.

2. Easily located.

3. Easily corrected by selective grinding.

4. The articulating paper marks can be quite easily made on dry teeth.

5. Make the correction away from the patient thus there is a psychological advantage.

Articulating paper and wax sheet can be used to detect the premature contacts, although it is preferable to use wax sheet because premature contact will cause the cusps to penetrate through the wax indicating heavy contact is present.

An alternate method is to correct the occlusion before removal the dentures from the cast, as follows:

1. Replace the upper and lower mountings casts and dentures on the articulator. If processing changes in occlusion have occurred. They must be corrected.

2. Re-established the vertical dimension of occlusion at this time, an opening in the vertical dimension can be noted by corresponding opening in the relationship of the incisal guide pin to the incisal table. The pin should contact the table.

3. Refine and equalize the centric occlusion.

4. Perfect the working and balancing occlusions.

5. Correct the protrusive occlusion.

Rules for selective grinding:

1. Never grind a centric cusp tip unless it contacts prematurely in all excursions of the mandible. Always grind the opposing fossa.

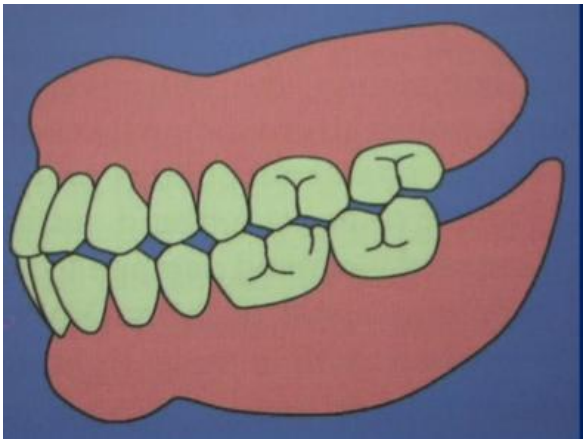
2. Utilize the **BULL** rule when perfecting **working occlusion** grind the **buccal cusp ridges of the upper and the lingual cusp ridges of the lower**.

3. When grinding to perfect **balancing occlusion** never grind the interfering cusp tips but grind the **cusp inclines**.

4. In correcting protrusive interference in the anterior teeth grind on the labial portion of the incisal edges of the lower teeth and the lingual portion (palatal surfaces) of the upper teeth.

For interference in the posterior teeth reduce the upper buccal cusp slopes and the lower lingual cusp slopes.

In protrusive balance, the anterior teeth should make incisal edge contact at the same time that the tips of the buccal and lingual cusps of the posterior teeth contact.



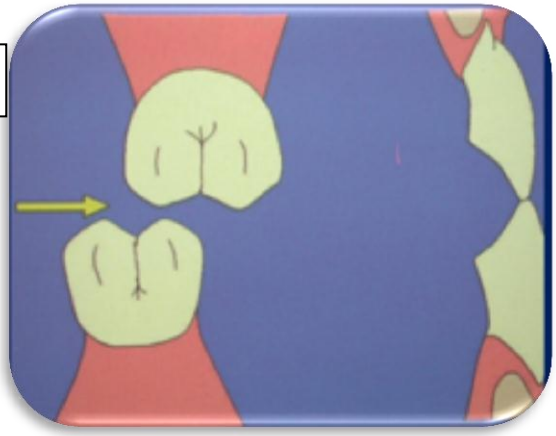
A. If the anterior teeth have heavy contact with no posterior contact:

Reduce the labio-incisal surfaces of the lower teeth and palatal surfaces of the upper teeth.

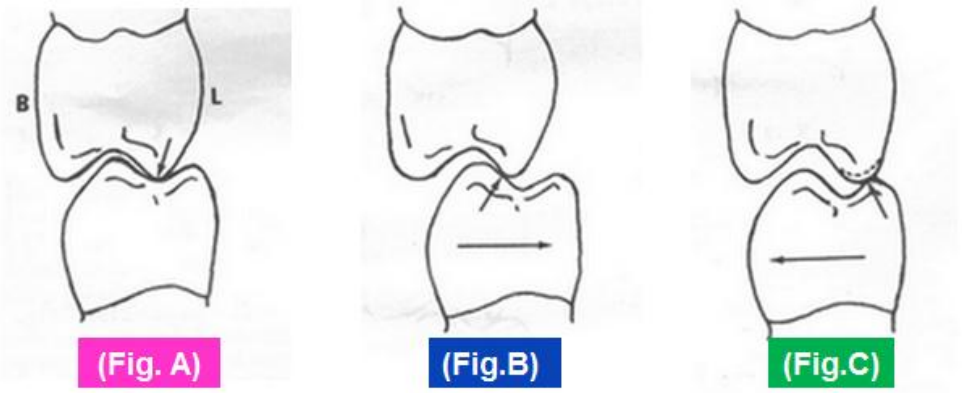
B. If the posterior teeth have heavy contact with no anterior teeth contact:

Reduce distal inclines of upper cusps and mesial inclines of lower cusps

A



If a cusp is in premature contact in centric relation (Fig. A) and is also premature in balancing (Fig.B) or working (Fig. C) then the cusp should be reduced (Fig.C).

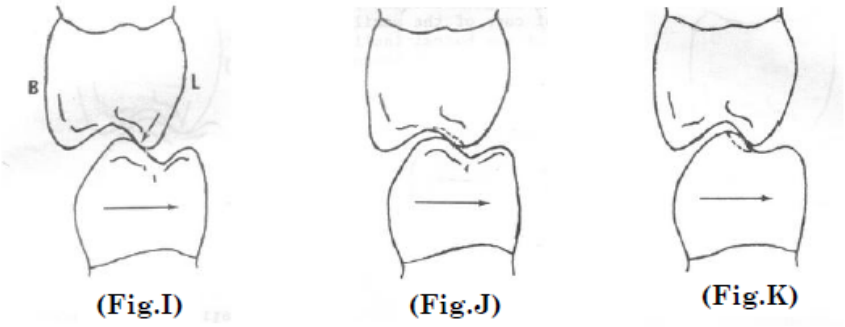


Centric prematurity

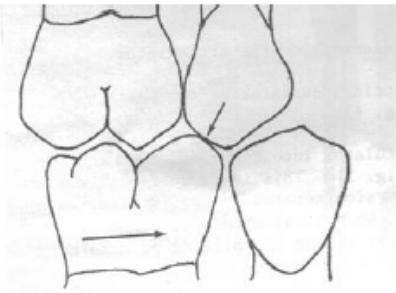
If the cusp is in premature contact in centric relation (Fig.D) but is not in working (Fig.E) or balancing (Fig.F) prematurity, then the opposing fossa or

When adjusting

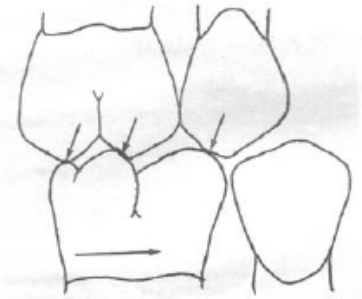
If there is a premature contact on the balancing side (Fig.I), then adjust the buccal incline plane of the lingual cusp of the maxillary tooth (Fig.J) or the lingual incline plane of the buccal cusp of the mandibular tooth (Fig.K). Do not adjust both upper and lower. Try not to adjust the tip of the cusp. Centric holding cusps must be maintained.



If there is a premature contact in protrusive excursion (Fig.L) , then desired contact (Fig.M) may be obtained by adjusting the distal cusp inclines of the upper teeth (Fig.N) or the mesial cusp inclines of the lower teeth.



(Fig.L)



(Fig.M)

(Fig.N)